

Chap 9 Strings

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Strings

■ Strings

- Strings are used for storing text/characters.
- For example, "Hello World" is a string of characters.
- Unlike many other programming languages, C does not have a **String type** to easily create string variables.
- Instead, you must use the char type and create an array of characters to make a string in C:

```
char greetings[] = "Hello World!";
```

- To output the string, you can use the printf() function together with the format specifier %s to tell C that we are now working with strings:

```
char greetings[] = "Hello World!";  
printf("%s", greetings);
```

Strings

■ Access Strings

- strings are actually arrays in C, you can access a string by referring to its index number inside square brackets []

```
char greetings[] = "Hello World!";  
printf("%c", greetings[0]);
```

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■ Modify Strings

- To change the value of a specific character in a string, refer to the index number, and use single quotes:

```
char greetings[] = "Hello World!";  
greetings[0] = 'J';  
printf("%s", greetings);  
// Outputs Jello World! instead of Hello World!
```

Strings

■ Loop Through a String

- You can also loop through the characters of a string, using a for loop:

```
char carName[] = "Volvo";
int i;

for (i = 0; i < 5; ++i) {
    printf("%c\n", carName[i]);
}
```

```
char carName[] = "Volvo";
int length = sizeof(carName) / sizeof(carName[0]);
int i;

for (i = 0; i < length; ++i) {
    printf("%c\n", carName[i]);
}
```

Strings

■ Differences

- The difference between the two ways of creating strings, is that the first method is easier to write, and you do not have to include the `\0` character, as C will do it for you.
- You should note that the size of both arrays is the same: They both have 13 characters (space also counts as a character by the way), including the `\0` character:

```
char greetings[] = {'H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', '!', '\0'};
char greetings2[] = "Hello World!";

printf("%zu\n", sizeof(greetings)); // Outputs 13
printf("%zu\n", sizeof(greetings2)); // Outputs 13
```

Strings Functions

■ String Functions

- C also has many useful string functions, which can be used to perform certain operations on strings.
- To use them, you must include the **<string.h>** header file in your program:

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main(){
5     char alphabet[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
6     printf("%u\n", strlen(alphabet));
7     return 0;
8 }
```

Strings Functions

■ *sizeof* vs *strlen*

- we used `sizeof` to get the size of a string/array. Note that `sizeof` and `strlen` behaves differently, as `sizeof` also includes the `\0` character when counting

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main(){
5     char hello[] = "Hello";
6     printf("strlen = %u\n", strlen(hello)); // 5
7     printf("sizeof = %u\n", sizeof(hello)); // 6
8     return 0;
9 }
10
```

Strings Functions

■ Concatenate Strings

- To concatenate (combine) two strings, you can use the `strcat()` function:
- `strcat( dest, source )`

```
1 #include <stdio.h>
2 #include <string.h>
3 // #define _CRT_SECURE_NO_WARNINGS
4
5 int main(){
6     char str1[20] = "Hello ";
7     char str2[] = "World!";
8     // Concatenate str2 to str1 (result is stored in str1)
9     strcat(str1, str2);
10    // Print str1
11    printf("%s\n", str1);
12    return 0;
13 }
```

Strings Functions

■ Copy Strings

- To copy the value of one string to another, you can use the strcpy() function:

```
1 #include <stdio.h>
2 #include <string.h>
3 // #define _CRT_SECURE_NO_WARNINGS
4
5 int main(){
6     char str1[20] = "Hello World!";
7     char str2[20];
8     // Copy str1 to str2
9     strcpy(str2, str1);
10    // Print str2
11    printf("%s\n", str2);
12    return 0;
13 }
```

Strings Functions

■ Compare Strings

- To compare two strings, you can use the strcmp() function.
- It returns 0 if the two strings are equal, otherwise a value that is not 0:

```
1 #include <stdio.h>
2 #include <string.h>
3 //define _CRT_SECURE_NO_WARNINGS
4
5 int main(){
6     char str1[] = "Hello";
7     char str2[] = "Hello";
8     char str3[] = "Hi";
9     // Compare str1 and str2, and print the result
10    printf("%d\n", strcmp(str1, str2)); // Returns 0 (the strings are equal)
11    // Compare str1 and str3, and print the result
12    printf("%d\n", strcmp(str1, str3)); // Returns -4 (the strings are not equal)
13    return 0;
14 }
15
```

Thank You !